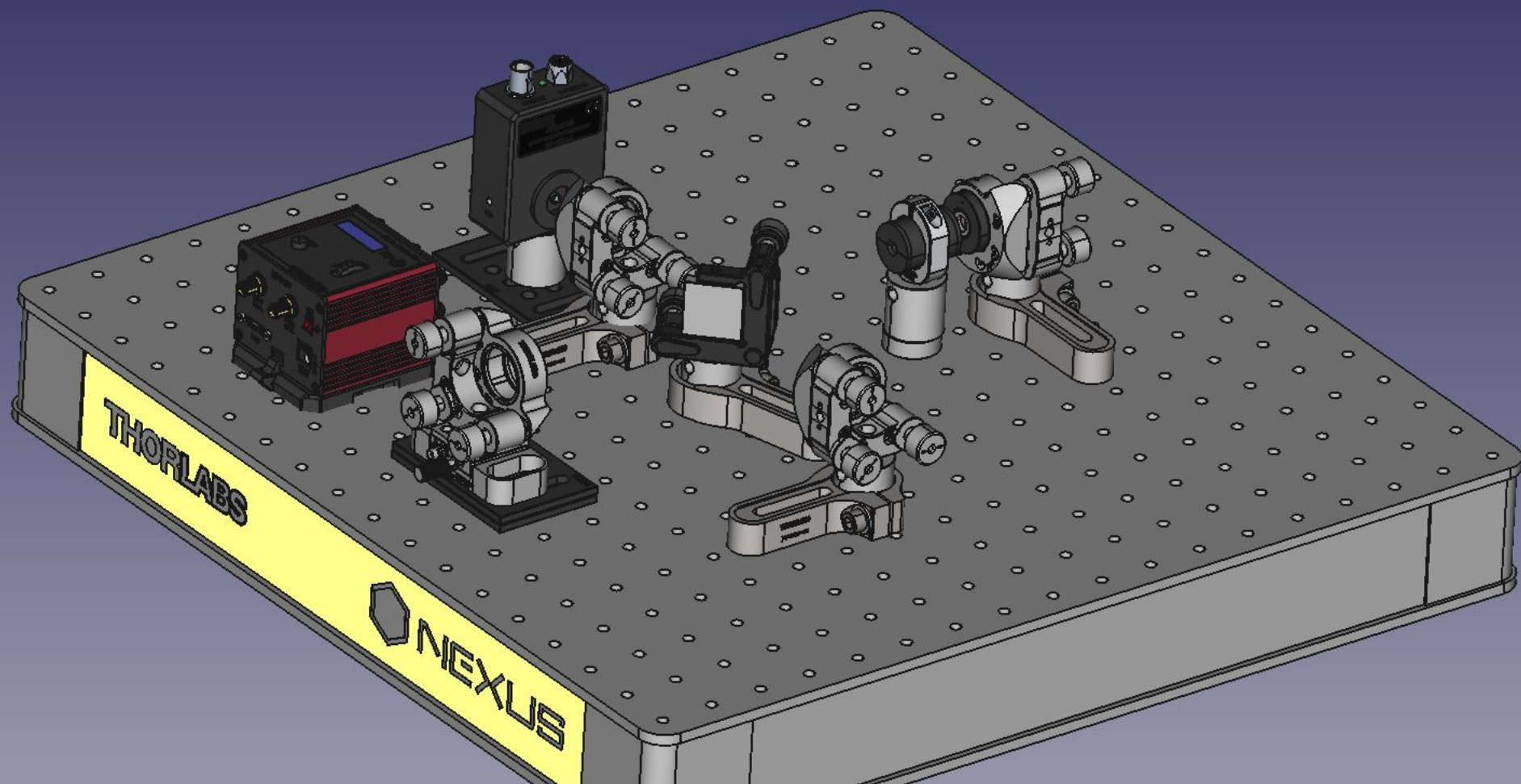
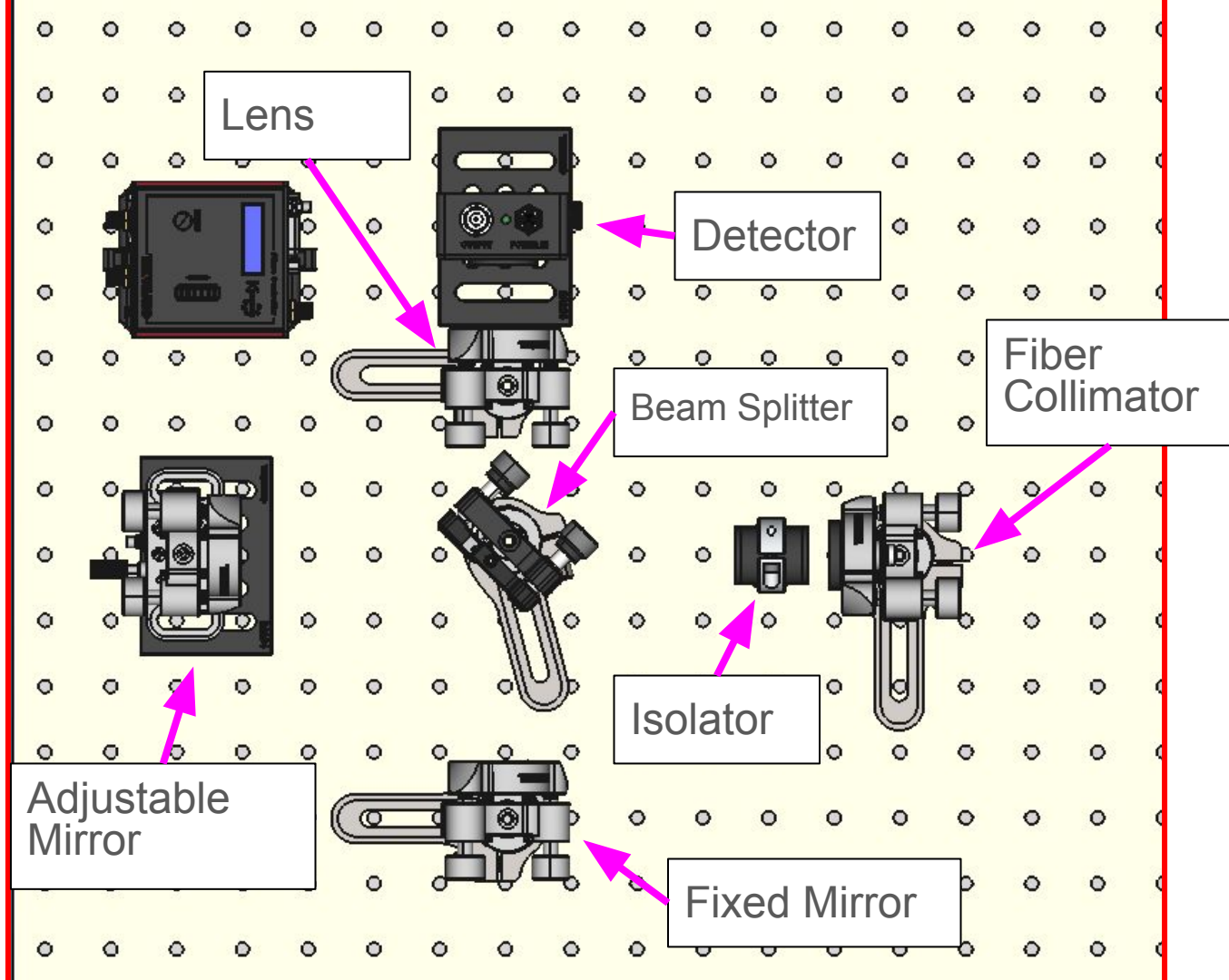


# Interferometer



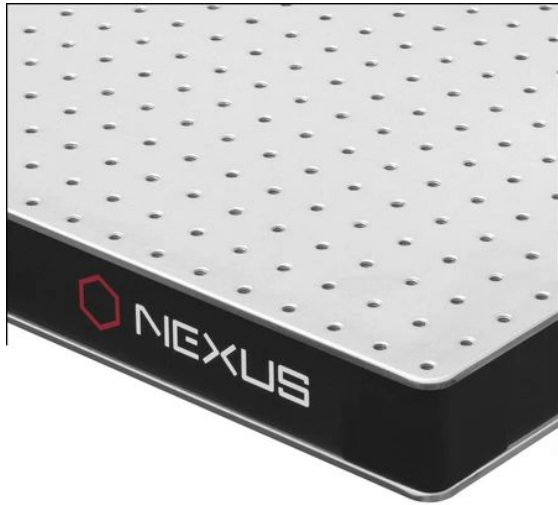


# Components - Base

Breadboard: 18x18x2.4

Thorlabs B1818F

Optics Height 2" (50mm)



# Components - Moveable Mirror

NF15P1 Mounting Plate



Polaris K1T1 Mirror Mount



NF15AP25NanoFlex™ 1.5 mm Travel Translation Stage with 25  $\mu$ m Piezo Actuator



KPC101 Controller



**KPC101**  
Shown with included  
Mounting Plate

# Components - Laser Collimator

AD12F



TC12APC-1550 1550 nm  
FC/APC Triplet Collimator



PLS-P1 1" post



Polaris CA1



Polaris K1T mount



# Components - Fixed Mirror and Lens

PLS-P1 1.5" post



Polaris CA1



Polaris K1T



# Components - Isolator

IOT-4-1550-VLP

Free-Space Tandem

Isolator, 1550 nm, Ø3.6 mm

Max Beam, 2.4 W Max



PLS-P1 1" post



Polaris CA1



# Components - Detector

APD410C

InGaAs Variable-Gain

Avalanche Photodetector,  
Temperature Compensated,  
900 - 1700 nm, DC - 10 MHz



# Questions

1. Collimator gives approx 2.5 mm beam size. Good?
2. Plate beam splitter ok? Better mounting?
3. Are posts, clamps, mounts overkill?
4. Detector has small area. Focusing ok?
5. Is isolator required?
6. Is mirror ok for fixed reflector? Retroreflector?

# Cost Breakdown

Current cart for approx \$11k

\$1k for breadboard

\$500 for hardware and tools

\$2k for positioning stage and controller

\$3k for detector

\$2k for isolator

\$2k for mounts